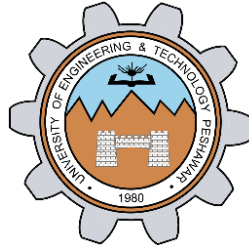


Introduction to 8051 Microcontroller & Installation of Software and Hardware Tools

LAB # 01



Spring 2024

CSE-307L Microprocessor Based System Design Lab

Submitted by: **Ali Asghar**

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Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Dr. Asif Ali Khan

Date:

25th February 2024

**Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar**

Task:

Write assembly and C language code to toggle the led attached with P1 and P2 continuously

Code:

```
Lab1.m51 task1_c.c task1.a51
1  #include<reg51.h>
2
3  int a;
4
5  void main(){
6
7      while(1){
8          a = 0xFF;
9          P1 = a;
10         P2 = a;
11         a = 0x00;
12         P1 = a;
13         P2 = a;
14     }
15 }
```

In C

```
Lab1.m51 task1_c.c task1.a51 lab2.a
1  ORG 0000h
2      HERE:
3      MOV A, #0FFH
4      MOV P1, A
5      MOV P2, A
6      MOV A, #00H
7      MOV P1, A
8      MOV P2, A
9      SJMP HERE
10
11  END
```

In Assembly

Output:

```
Lab1.m51 task1_c.c task1.a51
3      MOV A, #0FFH
4      MOV P1, A
5      MOV P2, A
6      MOV A, #00H
7      MOV P1, A
8      MOV P2, A
9      SJMP HERE
10
11  END
```

Parallel Port 2

Port 2	7	Bits	0
P2: 0xFF	☒	☒	☒
Pins: 0xFF	☒	☒	☒

Parallel Port 1

Port 1	7	Bits	0
P1: 0x00	☐	☐	☐
Pins: 0x00	☐	☐	☐

```
Lab1.m51 task1_c.c task1.a51
3      MOV A, #0FFH
4      MOV P1, A
5      MOV P2, A
6      MOV A, #00H
7      MOV P1, A
8      MOV P2, A
9      SJMP HERE
10
11  END
```

Parallel Port 2

Port 2	7	Bits	0
P2: 0x00	☐	☐	☐
Pins: 0x00	☐	☐	☐

Parallel Port 1

Port 1	7	Bits	0
P1: 0xFF	☒	☒	☒
Pins: 0xFF	☒	☒	☒

Proteus Simulation:

